BCSE XXX: DIGITAL FORENSIC & SOC AUDITING

Course Objectives: To equip students with the skills to identify, collect, analyze, and preserve digital evidence for effective cyber incident investigation and legal compliance. understanding of SOC auditing practices for monitoring security events, assessing incident response, and improving an organization's overall security posture.

Credits: 03 **L–T–P: 3–0–0**

Module No.	Contents	Hours
I	**Foundations of Digital Forensics:** Definition, Objectives, Types (Computer, Network, Mobile), Legal Considerations, Chain of Custody, Role of Investigators, Forensic Process (Identification, Collection, Preservation, Analysis, Reporting). **Evidence Collection & Preservation:** Types of Digital Evidence, Collection Methods, Imaging & Cloning (FTK, dd), Integrity Verification (Hashing: MD5, SHA-1, SHA-256), Evidence Handling & Storage. **System & Storage Analysis:** File Systems (FAT, NTFS, HFS+, ext), File Carving, Metadata & Slack Space Analysis, Partition Structures, Windows (Registry, Event Logs, Prefetch, Shortcuts) & Linux/Mac Forensics (Logs, Command History, File Permissions). **Data Recovery & Storage Forensics:** Deleted File Recovery, File Carving, Encrypted & Hidden Data Retrieval, Recovery Tools (FTK Imager, EnCase). **Malware & Threat Analysis:** Malware Types, Static & Dynamic Analysis, Sandboxing (Cuckoo), Reverse Engineering Basics (IDA Pro, Ghidra), Process Injection & Hidden Threats.	20
II	**SOC Fundamentals & Operations:** Introduction to Security Operations Center (SOC), SOC Roles & Responsibilities (Analyst L1, L2, L3), SOC Workflow: Monitoring/Detection/Response, SOC Tools Overview (SIEM, EDR, IDS/IPS). **Incident Detection & Response:** Incident Detection Techniques (Logs, Alerts, SIEM correlation), Incident Classification & Prioritization, Incident Response Lifecycle, Basics of Threat Intelligence & Use Cases. **Vulnerability & Risk Management:** Vulnerability Scanning (Nessus, OpenVAS – overview), Risk Identification & Prioritization, Patch Management Basics, Risk Mitigation Techniques. **Compliance & Security Standards:** Overview of Compliance: ISO 27001, NIST, PCI-DSS, Security Policies & Procedures, Basic Regulatory Requirements (BFSI, IT Act India – overview), Audit Readiness Concepts. **Security Auditing & Tools:** Introduction to Security Auditing, Audit Process & Checklist Creation, Log Analysis & Reporting, Tools Overview: Suricata, Wazuh, snort, Security Onion, Graylog Open, OpenSearch, OSSIM.	20

Reference Books:

- Digital Forensics and Incident Response – Kevin Mandia, Matthew Pepe, Jason Luttgens 1st Edition, 2014
- Guide to Computer Forensics and Investigations – Bill Nelson, Amelia Phillips, Christopher Steuart 6th Edition, 2018
- Practical Digital Forensics – Niranjan Reddy 1st Edition, 2016
- "Auditing Information Systems" by Jack J. Champlain (2022)
- "Information Security Governance: A Practical Development and Implementation Approach" by W. Krag Brotby (2021)
- "Information Security: Principles and Practice" by Mark Stamp and Mark M. Khosrowpour (3rd Edition, 2020)

- "ISACA's IT Audit Fundamentals" by ISACA (2020)

Focus: This course focuses on developing practical expertise in digital forensic investigation and SOC operations to handle real-world cyber incidents

Employability: This course enhances employability by providing hands-on forensic and SOC skills highly demanded in cybersecurity roles

- Digital Forensic Analyst
- SOC Analyst (L1 / L2)
- Incident Response Analyst
- Cyber Security Analyst
- Threat Intelligence Analyst
- Security Auditor / Compliance Analyst

Learning Outcomes: After completion of the course, student will be able to:

CO1: Understand the fundamentals of digital forensics, including evidence handling, legal considerations, and forensic processes
CO2: Perform evidence acquisition, imaging, and integrity verification using hashing techniques
CO3: Analyze file systems, logs, and system artifacts across Windows, Linux, and other platforms
CO4: Apply data recovery and forensic analysis techniques for deleted, hidden, and encrypted data
CO5: Evaluate malware behavior using static and dynamic analysis techniques
CO6: Understand SOC architecture, workflows, and tools for monitoring and incident detection
CO7: Analyze security events, perform incident response, and conduct basic security auditing and compliance checks

COs	POs/PSOs
CO1	PO1, PO2 / PSO1
CO2	PO2, PO3, PO4 / PSO2
CO3	PO1, PO3, PO5 / PSO3
CO4	PO2, PO4, PO6 / PSO4
CO5	PO3, PO5, PO7 / PSO2, PSO4
CO6	PO4, PO5, PO6 / PSO3
CO7	PO3, PO6, PO8 / PSO4

BCSE XXXX: DIGITAL FORENSIC & SOC AUDITING LAB

Course Objectives: To provide hands-on experience in performing digital forensic investigations, including evidence acquisition, analysis, and reporting using standard tools and techniques. To develop practical skills in SOC operations such as log monitoring, alert analysis, incident handling, and security auditing in a simulated environment.

Credits:01 **L-T-P:0--2**

Sr. No	LAB EXPERIMENTS	Tools
1	**Imaging and Hashing Techniques** **Objective**: Learn disk imaging and hashing for data integrity. Create a forensic image using FTK Imager and validate it with hashing (e.g., MD5, SHA-1).	FTK Imager
2	**File Metadata and Timestamps Analysis** **Objective**: Investigate file properties for evidence. Analyze metadata and timestamps of files using forensic tools.	Autopsy, ExifTool, Sleuth kit
3	**Deleted Partition Recovery** **Objective**: Recover deleted partitions and analyze their data. Use TestDisk or FTK Imager to recover and explore data from deleted partitions.	TestDisk or FTK Imager
4	**Windows Registry Forensics** **Objective**: Extract evidence from Windows Registry. Use tools like regedit and RECmd to analyze registry keys for user activity and system configuration changes.	regedit and RECmd
5	**Memory Data Collection and Analysis** **Objective:** Extract and analyze data from RAM. Use Volatility to extract running processes, open files, and network connections from memory dumps.	Volatility, FTK-Imager, WinPMEM, lime
6	**Linux Logs and User Activity Forensics** **Objective:** Extract and analyze Linux system logs. Explore syslog, auth.log, and bash history to track user activity and system events.	Syslog, journalctl, grep, awk, sed
7	**Wazuh Implementation** **Objective:** Set up Wazuh server, manager, and agents on Windows/Linux machines.	wazuh
8	**Splunk free** **Objective:** Deploy Splunk on Windows or Linux for local data analysis, Collect logs from system files, syslog, or Wazuh/Filebeat.	Splunk
9	**Nessus Essentials Vulnerability Scanner** **Objective:** Run basic network scans, host discovery, and credentialed scans on local VMs or lab targets.	Nessus
10	**Network Analysis Webshell** **Objective:** use Wireshark tool understand .pcap file to analysis webshell in network and identifies attack surface	Wireshark
11	**Network Analysis Ransomware** **Objective:** use Wireshark tool understand .pcap file to analysis Ransome attack in network and identifies attack surface	Wireshark
12	**Detecting Brute Force Attack Using SIEM** **Objective**: Simulate brute-force attacks and analyze detection using SIEM.	Hydra, Splunk/ELK
13	**Linux Syslog Monitoring** **Objective:** Configure and monitor /var/log files for suspicious activities.	Syslog, rsyslog, Graylog
14	**Sysinternals tools** **Objective:** Using Sysinternals tools is a powerful way to analyze and troubleshoot Windows systems or applications. These tools	Sysinternals Suite

15	**Lynis Security Audit Tool** **Objective**: Install the tool and perform a full system audit using command-line options.Analyze the audit report and understand the meaning of warnings, suggestions, and security index.	Lynis

Reference Books:

- *Digital Forensics and Incident Response* – Kevin Mandia, Matthew Pepe, Jason Luttgens (1st Edition, 2014)
- *Guide to Computer Forensics and Investigations* – Bill Nelson, Amelia Phillips, Christopher Steuart (6th Edition, 2018)
- *Practical Digital Forensics* – Niranjan Reddy (1st Edition, 2016)
- *Auditing Information Systems* – Jack J. Champlain (2022)
- *Information Security Governance* – W. Krag Brotby (2021)
- *Information Security: Principles and Practice* – Mark Stamp (3rd Edition, 2020)
- *ISACA IT Audit Fundamentals* – ISACA (2020)

Focus: The lab focuses on **hands-on digital forensic investigation and SOC operations**, enabling students to collect, analyze, and preserve digital evidence while performing real-time monitoring and incident response in simulated environments.

Employability: This lab equips students with practical skills required for high-demand cybersecurity roles:

- Digital Forensic Analyst
- SOC Analyst (L1 / L2)
- Incident Response Analyst
- Cyber Security Analyst
- Threat Intelligence Analyst
- Security Auditor / Compliance Analyst

Course Outcomes:

CO1: Perform digital evidence acquisition, imaging, and hashing to ensure data integrity.
CO2: Analyze file systems, metadata, and system artifacts for forensic investigation.
CO3: Conduct memory and log analysis to identify suspicious activities and user behavior.
CO4: Apply data recovery techniques to retrieve deleted, hidden, and encrypted information.
CO5: Evaluate malware and network-based attacks using forensic and analysis tools.
CO6: Implement SOC operations including log monitoring, SIEM analysis, and alert correlation.
CO7: Perform incident response and basic security auditing aligned with compliance standards.

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- Obtaining Protected System Files
- Detecting EFS Encryption
- Exporting Files from an Image

Step 3: Select the Specific Drive

From the drop-down menu, choose the exact drive you want to image.

Step 4: Add the Destination Path

Next, you'll need to tell FTK Imager where to save the new image.

Forensic Best Practice: Always save your forensic images to a separate, dedicated storage drive. It's also smart to create multiple copies of the original evidence to safeguard against accidental data loss or hardware failure.

Step 5: Select the Image Format

You will be prompted to choose a file format for your image. Here is the quick breakdown of your options:

- Raw (dd): A pure, bit-by-bit copy of the original evidence. No metadata is added, and nothing is deleted. It's universally compatible but lacks built-in verification details.
- SMART: An older image format primarily used for Linux systems, though it has mostly fallen out of common use.
- E01 (EnCase): The industry standard developed by EnCase. It compresses data and embeds metadata, case details, and hashes directly into the file.
- AFF (Advanced Forensic Format): An open-source extensible file format designed to store disk images and metadata.

Pick the format required for your investigation (E01 is usually the go-to default for modern workflows).

Step 6: Enter the Case Details

FTK Imager will ask for information like the Case Number, Evidence Number, Unique Description, and Examiner Name. Fill these out to keep your paperwork pristine.

Step 7: Name and Finalize

Browse to your target folder, name the image file, and click Finish.

Step 8: Start and Verify

Before you hit start, make sure the Verify images after they are created box is checked. This ensures the tool automatically calculates the hashes. Click Start and let FTK Imager do its thing.

Step 9: Review the Hash Results

Once the process finishes, a Hash Results box will pop up. It will display the MD5 and SHA1 hashes of the source vs. the image. If they match, your copy is verified! It will also report if it encountered any bad sectors on the drive.

Capturing Volatile Memory (RAM)

Memory capture involves dumping the live contents of a computer's RAM into a non-volatile file (like a .raw or .mem file) on an external device. This is crucial because RAM holds volatile data—like active network connections, unencrypted passwords, and running malware—that vanishes the moment a machine is turned off or rebooted. You have to be incredibly careful here, as any action you take on a live system alters the memory slightly.

Step 1: Access the Capture Tool

In the top menu, go to File > Capture Memory.

Step 2: Configure and Capture

Browse to select your destination path, give the destination file a clear name, and choose whether you want to include the pagefile. Click Capture Memory.

Step 3: Wait for completion

A progress bar will track the dump. Once it completes, you have a snapshot of the system's live memory ready for analysis.

Analyzing your Image Dump

Once you have successfully captured a dump or an image, it's time to take a look inside.

Step 1: Load the Evidence

Go to File > Add Evidence Item...

Step 2: Choose the Source Type

Select Image File (since we are analyzing a dump we already created) and click Next.

Step 3: Browse to the File

Click Browse, find your image dump file, and click Finish.

Step 4: Explore the Evidence Tree

Look at the left-hand pane: you will see the Evidence Tree. You can expand these folders to browse the filesystem structure. FTK Imager will display active files, deleted files (marked with a red 'X'), and potentially overwritten data.

Step 5: Cleaning Up the Workspace

When you are done exploring a specific piece of evidence, you can tidy up your workspace. Right-click on the case or image name in the Evidence Tree and select Remove Evidence Item. (Don't worry, this doesn't delete the actual file on your hard drive; it just removes it from the view).

Mounting an Image as a Drive

Sometimes, you need to view a forensic image as if it were a regular, physical drive plugged into your machine. FTK Imager makes this seamless.

Step 1: Open Image Mounting

Go to File > Image Mounting...

Step 2: Select the File and Mount

In the Mount Image window, browse to your image file. You can choose a specific drive letter and mount method (like "Read-Only," which is critical for maintaining forensic integrity). Click Mount.

Your operating system will now treat that image just like an attached external drive, letting you browse it natively through Windows File Explorer.

Creating Encrypted Custom Content Images

If you don't need a whole drive but just want to extract specific, highly sensitive files, you can create a Custom Content Image and protect it with AccessData (AD) encryption.

Step 1: Select Your Target Files

Right-click on the specific files or folders inside the FTK Imager file view that you want to isolate, and choose to add them to your custom content list.

Step 2: Initiate Image Creation

A window will display all your selected items. Review them, then click Create Image.

Step 3: Add Case Info & Choose Destination

Fill out the standard evidence details, click next, and select your destination path. Make sure to check the box for AD Encryption, then click Finish.

Step 4: Set the Password

A password prompt will appear. Type in a strong password, re-enter it to confirm, and click OK to generate the encrypted file.

Step 5: Viewing the Encrypted Content

If you want to view this custom image later, load it via File > Add Evidence Item... As soon as you select the file source, a prompt will challenge you for the password. Enter it, click OK, and your custom files will be visible in the tree.

Decrypting AD1 Images

If you want to permanently strip the password encryption off a custom content image (.ad1 file) to share it with other team members or tools, you can decrypt it directly.

Step 1: Select Decrypt Option

Go to File > Decrypt AD1 Image...

Step 2: Provide the Credentials

Browse to the encrypted image file, provide the correct password, and choose a new destination folder for the output. Click OK.

The tool will process the file for a few moments, creating a fully unencrypted version of the custom content image at your specified destination. You can now load it into any analysis tool without a password prompt.

Obtaining Protected System Files

Windows locks down critical registry hives and system security files while the OS is running, making them tough to copy normally. FTK Imager allows you to extract these protected files directly from a live system or an image.

Step 1: Select Obtain Protected Files

Navigate to File > Obtain Protected Files...

Step 2: Choose Extraction Type

In the menu that pops up, select your destination directory. You can choose to extract just the minimum registry files needed for password recovery, or grab All registry files (SAM, SYSTEM, SECURITY, SOFTWARE, etc.). Click OK.

FTK Imager will bypass the OS locks and export these vital configuration files into your destination folder, ready to be cracked or audited.

Detecting EFS Encryption

The Encrypting File System (EFS) is a built-in Windows feature that users leverage to protect specific files and folders. If you are examining a drive, you need to know which files are encrypted so you can plan how to acquire the decryption keys.

Step 1: Scan for Encryption

With your evidence item loaded into the tree view, go to File > Detect EFS Encryption.

Step 2: Review Findings

The tool will scan the directories and quickly flag any files utilizing EFS. This alerts you right away to the presence of protected data that might otherwise surprise you during deeper analysis.

Exporting Files from an Image

When you find a file or folder inside a forensic image that is highly relevant to your case, you will want to pull it out into the real world for further study or reporting.

Step 1: Run the Export

Right-click on the desired file or folder within the Evidence Tree pane, or go to File > Export Files...

Step 2: Save and Confirm

Choose the folder on your analyst workstation where you want to save the files. Once the export completes, a summary window will pop up showing you exactly how many files and folders were successfully copied out of the image.